

RETAIL SALES OPTIMIZATION

Data Analytics & Business Intelligence Project

Prepared By

Muhammad Salman

Tools & Technology

- Python
- Pandas
- NumPy
- MySQL
- SQL
- Power BI
- Matplotlib
- MySQL ODBC Connector

Executive Summary

Retail businesses generate large volumes of sales data every day. The objective of this project was to analyze retail sales transactions, identify customer purchasing patterns, evaluate product performance, and generate actionable business insights that can improve revenue and sales performance.

The project involved:

- Data Cleaning and Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- SQL-Based Business Analysis
- Interactive Dashboard Development in Power BI
- Business Recommendation Generation

The final outcome provides a complete analytical solution that helps management understand customer behavior, product demand, seasonal trends, and sales opportunities.

Business Problem

Retail organizations need to answer several critical questions:

- Which product category generates the highest revenue?
- Which customer segment contributes the most sales?
- Which months generate the highest sales?
- What purchasing patterns exist among different age groups?
- How can sales performance be improved through data-driven decisions?

This project aims to answer these questions using historical sales data.

Dataset Overview

The dataset contains retail transaction records with customer demographics and purchasing information.

Dataset Features

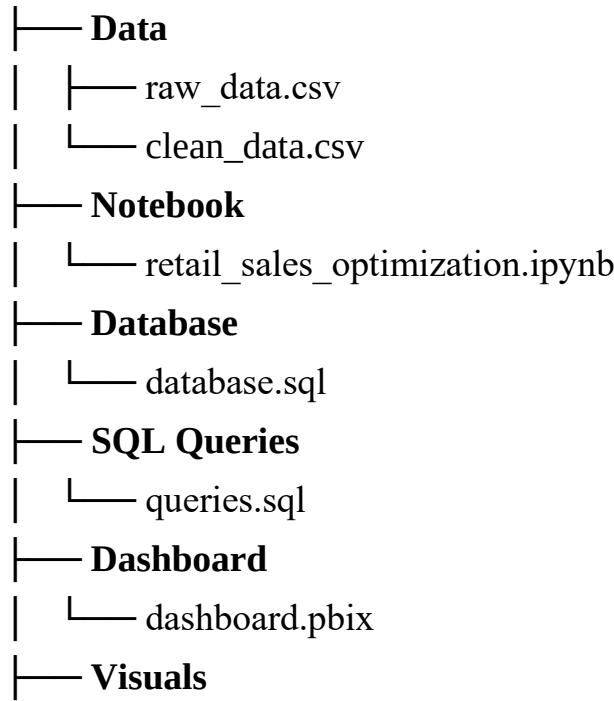
Column Name	Description
Transaction ID	Unique transaction identifier
Date	Transaction date
Customer ID	Unique customer identifier
Gender	Customer gender
Age	Customer age
Product Category	Purchased product category
Quantity	Number of products purchased
Price per Unit	Unit price of product
Total Amount	Total transaction value

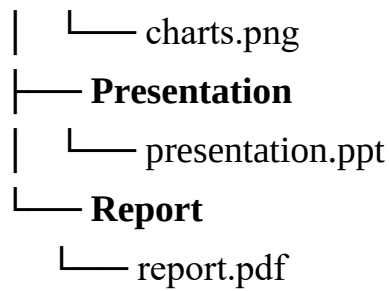
Dataset Size

- Total Records: 1,012
- Total Features: 9

Project Architecture

Retail Sales Optimization





Data Cleaning & Preprocessing

The raw dataset contained missing values and duplicate records that could affect analysis accuracy.

Missing Value Treatment

- Missing values in Gender column were replaced using Mode.
- Missing values in Product Category column were replaced using Mode.
- Total Amount column contained missing values and was reconstructed using:

Total Amount = Quantity × Price per Unit

Data Type Conversion

- Date column converted to DateTime format.

Duplicate Handling

- Identified duplicate records.
- Removed duplicate transactions to ensure data integrity.

Result

A clean and reliable dataset was prepared for further analysis.

Feature Engineering

Additional business-friendly features were created to improve analysis.

Age Category

- Teen
- Young
- Adult
- Middle Age
- Senior

Month

Extracted from transaction date for monthly trend analysis.

Year

Extracted from transaction date for yearly comparison.

Recalculated Total Amount

Price per Unit × Quantity

Exploratory Data Analysis (EDA)

Several analyses were performed to understand sales performance and customer behavior.

Analysis 1: Product Category Revenue

Objective

Identify the category generating maximum revenue.

Key Finding

Product Category	Revenue
Electronics	156,905
Clothing	155,580
Beauty	143,515

Insight

Electronics generated the highest overall revenue.

Analysis 2: Product Category Quantity Sold

Objective

Identify the category with highest sales volume.

Key Finding

Product Category	Quantity
Clothing	894
Electronics	849
Beauty	771

Insight

Clothing products achieved the highest sales volume

Analysis 3: Gender Purchasing Behavior

Objective

Compare spending behavior between genders.

Key Finding

Gender	Revenue
Female	233,740
Male	222,260

Insight

Female customers generated slightly higher revenue than male customers.

Analysis 4: Yearly Sales Performance

Objective

Evaluate annual sales trends.

Key Finding

Year	Revenue
2023	454,470
2024	1,530

Insight

Most transactions occurred during 2023.

Analysis 5: Monthly Sales Trend

Objective

Identify peak sales months.

Key Finding

Top Performing Months:

- 1. May
- 2. October
- 3. December

Insight

Sales tend to peak during specific seasonal periods.

Analysis 6: Age Category Performance

Objective

Identify the most valuable customer segment.

Key Finding

Age Category	Revenue
Middle Age	147,875
Adult	141,955
Young	121,730
Senior	33,225
Teen	11,215

Insight

Middle-aged customers are the most profitable customer group.

SQL Business Analysis

The cleaned dataset was loaded into MySQL for advanced business analysis.

Database Name

retail_sales_optimization

Table Name

sales_data

The following SQL analyses were performed:

- Category Revenue Analysis
- Category Quantity Analysis
- Gender-Based Revenue Analysis
- Yearly Sales Analysis
- Monthly Sales Analysis
- Age Category Analysis
- Product Preference by Age Group
- Revenue vs Quantity Comparison

SQL Query Results

Query 1: Highest Revenue Category

	Product Category	Revenue
►	Electronics	156905
	Clothing	155580
	Beauty	143515

Query 2: Highest Quantity Category

	Product Category	Quantity
▶	Clothing	894
	Electronics	849
	Beauty	771

Query 3: Gender Revenue Analysis

	Gender	Revenue
▶	Female	233740
	Male	222260

Query 4: Yearly Revenue Analysis

	Month	Revenue
▶	May	53150
	October	46580
	December	44690
	February	44060
	January	36980
	August	36960
	June	36715
	July	35465
	November	34920
	April	33870
	March	28990
	September	23620

Query 5: Monthly Revenue Analysis

	Age Category	Revenue
▶	Middle Age	147875
	Adult	141955
	Young	121730
	Senior	33225
	Teen	11215

Query 6: Age Category Revenue Analysis

	Age Category	Product Category	Revenue	quantity
▶	Middle Age	Electronics	51920	288
	Adult	Electronics	49375	259
	Middle Age	Clothing	48415	291
	Middle Age	Beauty	47540	252
	Adult	Clothing	47365	296
	Adult	Beauty	45215	218
	Young	Clothing	44585	205
	Young	Beauty	40750	222
	Young	Electronics	36395	204
	Senior	Electronics	17045	87
	Senior	Clothing	11130	83
	Senior	Beauty	5050	55
	Teen	Beauty	4960	24
	Teen	Clothing	4085	19
	Teen	Electronics	2170	11

Query 7: Age Group Product Preference

	Product Category	Revenue	Quantity
▶	Electronics	156905	849
	Clothing	155580	894
	Beauty	143515	771

Query 8: Revenue vs Quantity Analysis

	Year	revenue
▶	2023	454470
	2024	1530

Power BI Dashboard

An interactive Power BI dashboard was developed using data stored in MySQL.

Dashboard Components

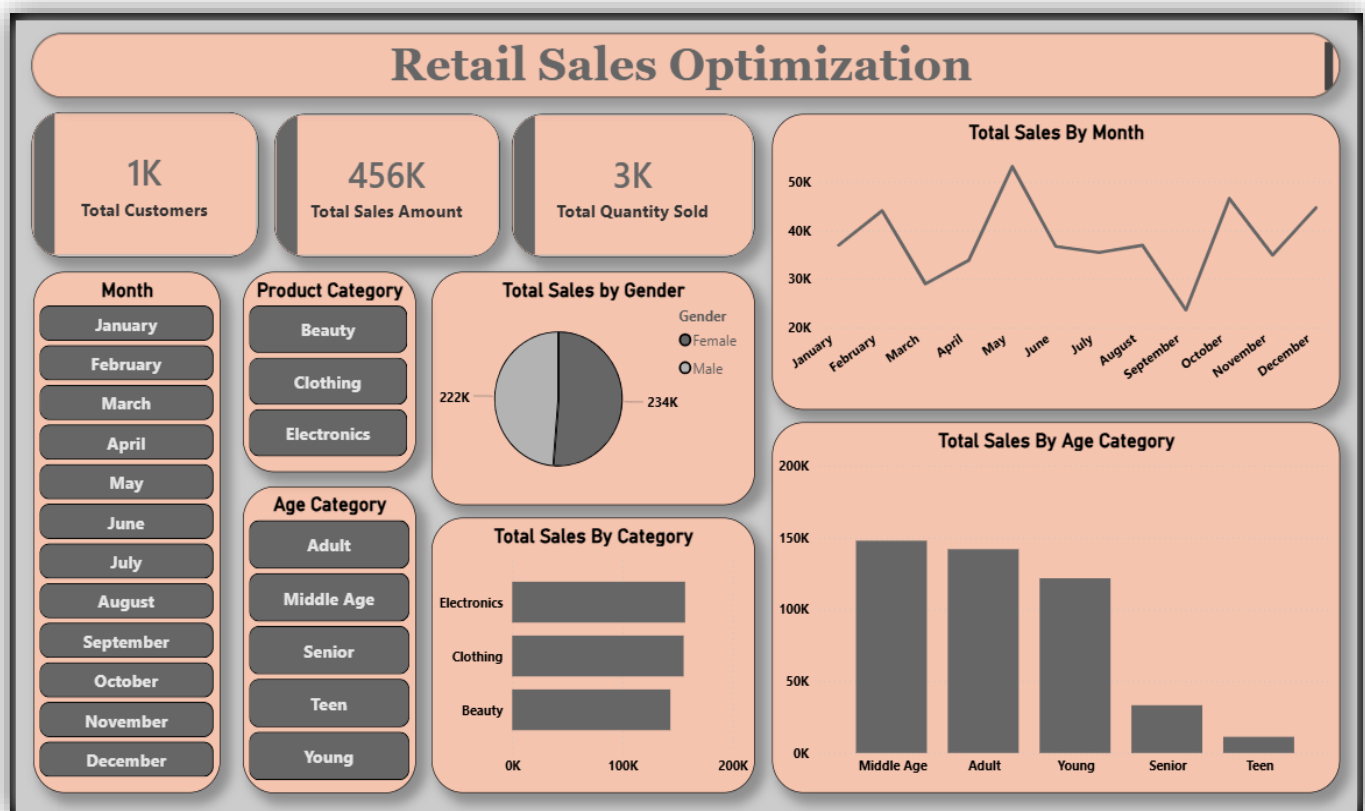
Slicers

- Month
- Product Category
- Age Category

Visualizations

- Pie Chart: Sales by Gender
- Horizontal Bar Chart: Sales by Product Category
- Vertical Bar Chart: Sales by Age Category
- Line Chart: Monthly Sales Trend

The dashboard enables dynamic filtering and interactive exploration of business performance.



Key Business Insights

- Electronics generates the highest revenue despite not having the highest quantity sold.
- Clothing products achieve the highest sales volume.
- Female customers contribute slightly more revenue than male customers.
- Middle-aged customers are the most profitable customer segment.
- May, October, and December are peak-performing months.
- Electronics products generate higher revenue per transaction compared to other categories.

Business Recommendations

Recommendation 1: Increase Electronics Promotions

Since Electronics contributes the highest revenue, the business should:

Suggested Actions:

- Increase marketing budget for electronics products.
- Launch seasonal electronics campaigns.
- Offer bundle deals on high-value products.

Expected Impact:

- Higher revenue growth.

Recommendation 2: Focus on Middle-Aged Customers

Middle-aged customers generate the highest sales.

Suggested Actions:

- Loyalty programs.
- Personalized offers.
- Targeted advertisements.

Expected Impact:

- Increased customer retention.

Recommendation 3: Optimize Inventory for Peak Months

Peak months include:

- May
- October
- December

Suggested Actions:

- Increase stock levels before peak periods.
- Improve supply chain planning.

Expected Impact:

- Reduced stock-outs and missed sales opportunities.

Recommendation 4: Boost Beauty Category Sales

Beauty category revenue is lower than Electronics and Clothing.

Suggested Actions:

- Product bundles.
- Promotional discounts.
- Cross-selling strategies.

Expected Impact:

- Improved category performance.

Recommendation 5: Improve Customer Segmentation

Develop separate marketing strategies for:

- Teens
- Young Adults
- Adults
- Middle-Aged Customers
- Seniors

Expected Impact:

- Higher conversion rates.

Recommendation 6: Implement Customer Loyalty Program

Reward repeat customers through:

- Membership discounts.
- Reward points.
- Exclusive offers.

Expected Impact:

- Increased customer lifetime value.

Conclusion

This Retail Sales Optimization project successfully transformed raw retail transaction data into actionable business intelligence. Through Python, SQL, and Power BI, valuable insights were generated regarding customer behavior, product performance, and sales trends.

The analysis identified Electronics as the leading revenue-generating category, Middle-Aged customers as the most profitable segment, and May, October, and December as peak sales periods. The Power BI dashboard provides an interactive decision-support system that enables management to monitor performance and make data-driven business decisions.